Comprehensive UX Benchmarking Report

Course: User Experience Design & Human-Computer Interaction (ICC612 / ICC716) — UFRO 2026-2
Initiative 03: Assessment Feedback that Students Actually Use (*LoopFeedback*)
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Deliverable: Milestone 1 (Hito 1) — Strategy & Scope Layer Artifact
Academic Reference: Adapted from the *Fructus Iter* benchmarking methodology.

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1. Initial Exploration Protocol (Pauta de Exploración Inicial)

1.1 Objective & Evaluation Framework

The goal of this benchmarking exploration is to evaluate how current educational and productivity platforms handle the transition from **evaluative feedback** (grades, rubrics, comments) to **actionable remediation** (student revision, evidence submission, and instructor verification).

We evaluate each platform across four standardized exploratory dimensions:

1. **D1: Criterion Granularity & Visibility:** Can the learner isolate the exact rubric criterion that failed without reading global unstructured text?
2. **D2: Action Prescriptiveness:** Does the platform translate descriptive critique ("unclear methodology") into a concrete corrective task?
3. **D3: Corrective Evidence Binding:** Can the learner attach and anchor proof of revision (code snippet, revised paragraph, screenshot) directly to the critique?
4. **D4: Longitudinal Traceability:** Can both student and instructor compare criterion performance across consecutive iterations side-by-side?

1.2 Standardized Exploratory User Tasks (Pauta de Tareas)

During the exploratory audit, the analyst executed four standardized test tasks under a shared illustrative scenario:

> *Scenario: Valentina received a low mark ($1/4$) on Criterion 3: "Justification of Design Decisions" in Milestone 1. She must identify what to fix, apply the correction, and prove to instructor Martina that the deficiency was resolved in Milestone 2.*

- **Task 1 (Locate & Interpret):** Access the returned deliverable, open the evaluation rubric, and identify the rationale for the penalty on Criterion 3.

- **Task 2 (Formulate Action):** Determine within the interface what specific step must be executed to achieve $4/4$ on the subsequent submission.
- **Task 3 (Submit Evidence):** Upload the revised version and explicitly link the modified section to the specific feedback comment.
- **Task 4 (Verification & Comparison):** As an instructor, review the new submission and verify whether the student corrected Criterion 3 without opening external reference documents.

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2. Competitive Reference Rationale

Platform	Category	Strategic Rationale for Inclusion
Moodle	Direct Academic Competitor	Institutional LMS standard at UFRO. Employs traditional grid-based rubrics. Baseline for institutional user mental models.
Google Classroom	Direct Academic Competitor	Widely used secondary/higher education platform. Features clean rubric scoring chips and simplified assignment workflows.
Canvas LMS	Direct Academic Competitor	Industry benchmark for modern higher education LMS. Features SpeedGrader with inline criterion comments.
Trello	Analogous Productivity Reference	Leading task-tracking tool. Demonstrates optimal interaction patterns for action checklists, status progression, and attached evidence.
GitHub (Pull Requests)	Analogous Collaborative Reference	Code review benchmark. Explores line-specific commentary, discussion resolution threads, and re-requesting review upon committing changes.
Figma (Canvas Comments)	Analogous Collaborative Reference	Design review benchmark. Demonstrates pinned in-situ contextual feedback and version comparison history.

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3. Annotated Interface Teardowns (Capturas Anotadas)

3.1 Moodle (Institutional LMS Baseline)

- **Analyzed Interface:** *Assignment Grading Sheet & Rubric Dialog View.*
- **Documented Support:** Criteria rows, discrete performance level cells, instructor free-text feedback box per criterion [S1].

Annotated Observations (Teardown Callouts):

- **[+] Callout A1 (Level Highlighting):** Selected rubric level is highlighted in light green, giving immediate visual feedback of achieved points.
- **[-] Callout A2 (Dispersed Comments):** If the instructor enters remarks, they appear inside a tiny, scrollable text area inside the grid cell, breaking visual hierarchy.
- **[-] Callout A3 (Zero Action Support):** Moodle provides no workflow to convert an assessment row into a student task. Feedback remains static archival text.
- **[-] Callout A4 (The Resubmission Void):** In the next submission, Moodle treats the new file as a complete overwrite. Previous rubric notes are buried in past submission logs.

3.2 Google Classroom (Simplified Rubric Chips)

- **Analyzed Interface:** *Student Returned Work & Graded Rubric Drawer.*
- **Documented Support:** Expandable rubric accordions with level pill chips and total point breakdown [S2].

Annotated Observations (Teardown Callouts):

- **[+] Callout B1 (Clean Visual Chips):** Each criterion is displayed as an expandable card with clear point pill badges, minimizing initial cognitive load.
- **[+] Callout B2 (Direct Rubric Visibility):** Students can access rubric levels before and after submission directly from the assignment stream.
- **[-] Callout B3 (Disconnected Private Comments):** Written feedback is relegated to a generic "Private comments" thread on the sidebar, detached from the individual rubric criterion.
- **[-] Callout B4 (Blind Resubmission):** Clicking "Resubmit" allows attaching a new Google Doc, but there is no mechanism to point the instructor to what changed.

3.3 Canvas LMS (SpeedGrader & Criterion Remarks)

- **Analyzed Interface:** *SpeedGrader Assessment Sidebar & Student Rubric Detail View.*
- **Documented Support:** Interactive rubric overlay, criterion-level comment bubbles, total

score computation [S3].

Annotated Observations (Teardown Callouts):

- **[+] Callout C1 (Criterion-Level Association):** Remarks are anchored directly beneath the specific criterion, resolving ambiguity about what the comment applies to.
- **[+] Callout C2 (Free-Form Media Feedback):** Allows text, audio, and video feedback per criterion, enriching formative guidance.
- **[-] Callout C3 (One-Way Communication):** The rubric is strictly an instructor-to-student broadcast. The student cannot reply or flag a criterion as resolved.
- **[-] Callout C4 (Absence of Cross-Submission Comparison):** Canvas does not support side-by-side criterion evolution across iterative submissions (e.g., Deliverable 1 vs. Deliverable 2).

3.4 Trello (Analogous Reference: Action & Evidence Tracking)

- **Analyzed Interface:** *Card Detail Modal with Checklists and File Attachments.*
- **Documented Support:** Item checklists, attachment preview, activity logs, member assignment [S4, S5].

Annotated Observations (Teardown Callouts):

- **[+] Callout D1 (Actionable Checklist Pattern):** Large, complex tasks are broken down into discrete checkable micro-actions with clear completion states.
- **[+] Callout D2 (Direct Evidence Attachment):** Users can attach screenshots, URLs, or documents directly to the card as verifiable proof of work.
- **[+] Callout D3 (Transparent Lifecycle):** Cards move predictably through workflow lists (`To Do` $\rightarrow$ `Doing` $\rightarrow$ `Done`).
- **[-] Callout D4 (Lack of Pedagogical Context):** No built-in concept of evaluation rubrics, grade weights, or academic assessor roles.

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4. Synthetic Comparison Matrix & ERIC Opportunity Matrix

4.1 Synthetic UX Comparison Matrix

*Legend: **D** = Documented native support; **A** = Manual adaptation proposed/observed; **NV** = Not verified / Absent in reviewed workflow.*

User Need / Evaluated Criterion	Moodle [S1]	Google Classroom [S2]	Canvas LMS [S3]	Trello [S4, S5]	LoopFeedback (Our Target)
1. Identify weak rubric criterion	**D** (Grid cell)	**D** (Card chip)	**D** (Rubric row)	**A** (Copy to card)	**D** (Ranked priority view)
2. Anchor explanation to criterion	**D** (Cell box)	**NV** (Global thread)	**D** (Comment box)	**A** (Card description)	**D** (Prescriptive Action Card)
3. Prescribe concrete remediation	**NV** (Prose only)	**NV** (Prose only)	**NV** (Prose only)	**D** (Checklist items)	**D** (Instructor action prompts)
4. Bind corrective evidence to critique	**NV**	**NV**	**NV**	**D** (Card attachments)	**D** (Deep-linked evidence uploader)
5. 3-Stage review status	**NV**	**NV**	**NV**	**A** (Board columns)	**D** (`Pending` $\rightarrow$ `Submitted` $\rightarrow$

User Need / Evaluated Criterion	Moodle [S1]	Google Classroom [S2]	Canvas LMS [S3]	Trello [S4, S5]	LoopFeedback (Our Target)
					$ `Approved`)
6. Longitudinal criterion comparison	**NV**	**NV**	**NV**	**A** (Manual history)	**D** (Sub 1 vs Sub 2 comparative diff)

4.2 ERIC Opportunity Matrix (Blue Ocean Framework)

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| ELIMINATE                                    | REDUCE
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| - Unstructured, multi-channel feedback dumps| - Time spent searching across
PDFs  |

| - "Read" status as proof of comprehension   | - Repetitive side-by-side reading
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| - Superficial checkbox completion claims    | - Cognitive overload of unranked
fix|

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| RAISE (INCREASE)                             | CREATE (INCLUDE)
|
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| - Transparency of rubric criteria & weights | - Prescriptive Action Cards
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| - Visibility of urgent corrective priorities| - Bidirectional Evidence Binding
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| - Clarity of instructor validation status   | - Longitudinal Criterion Diff
Viewer|

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5. Ideation & Concept Map (Mapa de Ideas)

The diagram below maps how the competitive benchmark directly informs the product architecture and user interface modules of **LoopFeedback**:

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flowchart TD

    subgraph BenchmarkInsights["Benchmark Insights (Sources)"]

        Moodle["Moodle & Canvas LMS:<br>Rubric Criteria & Levels"]

        Classroom["Google Classroom:<br>Scoring Chips & Clean Cards"]

        Trello["Trello:<br>Checklists & File Attachments"]

        Interviews["Empirical Fieldwork:<br>Need for Verifiable Proof"]

    end



    subgraph CoreComponents["LoopFeedback Product Components"]

        UI_Breakdown["1. Interactive Rubric Breakdown<br>(Clean cards, point
weights, status badges)"]

        UI_ActionCard["2. Prescriptive Action Cards<br>(Concrete steps to fix
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each penalized criterion)"]

        UI_Evidence["3. Evidence-Binding Module<br>(Attach snippet, URL or
excerpt directly to action)"]

        UI_Queue["4. Instructor Verification Queue<br>(Compare comment vs
evidence with 1-click approval)"]

    end



    subgraph ScopeAlignment["Milestone 1 Scope Alignment"]

        MainFlow["Core Flow:<br>Identify Penalty -> Read Action -> Bind
Evidence"]

        SecFlow1["Secondary Flow 1:<br>Instructor Verification & Approval"]

        SecFlow2["Secondary Flow 2:<br>Longitudinal Criterion History"]

    end



    Moodle --> UI_Breakdown

    Classroom --> UI_Breakdown

    Trello --> UI_ActionCard

    Trello --> UI_Evidence

    Interviews --> UI_Evidence

    Interviews --> UI_Queue



    UI_Breakdown --> MainFlow
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    UI_ActionCard --> MainFlow

    UI_Evidence --> MainFlow

    UI_Queue --> SecFlow1

    UI_Breakdown --> SecFlow2
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6. Official References & Evidence Documentation

- **[S1] MoodleDocs:** *Rubrics grading method and criterion level configuration.* (docs.moodle.org)
- **[S2] Google Classroom Support:** *Check your work and grades with rubrics.* (support.google.com)
- **[S3] Canvas LMS Community:** *How do I view rubric results and comments for my assignment?* (community.canvaslms.com)
- **[S4] Atlassian Trello Guide:** *Managing checklists and progressive task items.* (trello.com)
- **[S5] Atlassian Trello Guide:** *Adding attachments and visual proofs to cards.* (trello.com)